



# Chapter 8. Equilibrium Electrochemistry



# Outline

## The thermodynamic properties of ions in solution

- ✓ Thermodynamic functions of formation
- ✓ Ion activities

## Electrochemical cells

- ✓ Half-reactions and electrodes
- ✓ Varieties of cells
- ✓ electromotive force

## Applications of standard potentials



## 8.1 The thermodynamic properties of ions in solution

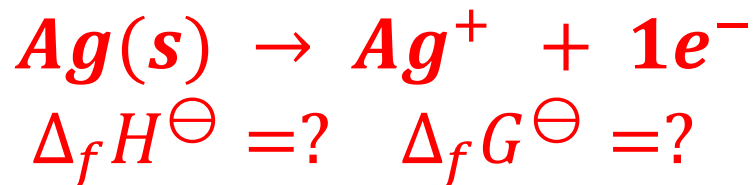
### 1. Thermodynamic functions of formation

#### (a) Standard functions of formation of ions

Enthalpy and Gibbs energy of formation values of  $\Delta_f H^\ominus$  and  $\Delta_f G^\ominus$  refer to formation of ions from reference state of parent elements.

Individual enthalpies and Gibbs energies for ions not directly measurable.

The following reaction cannot take place independently:





Only measure overall reactions

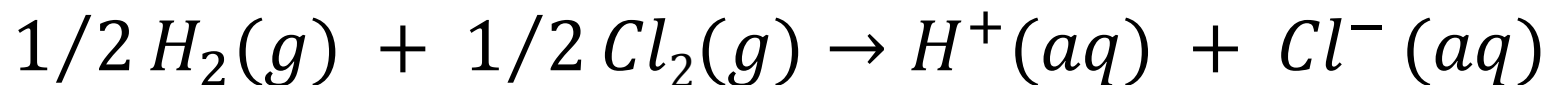


$$\begin{aligned}\Delta_r H^\ominus &= \Delta_f H^\ominus (Ag^+, aq) + \Delta_f H^\ominus (Cl^-, aq) \\ &= -61.58 \text{ kJ mol}^{-1}\end{aligned}$$

The  $\Delta_f H^\ominus$  of individual formation reactions are not measurable.

The problem is solved by defining one ion, conventionally the hydrogen ion, to have zero standard enthalpy and Gibbs energy of formation at all temperatures:

$$\Delta_f H^\ominus (H^+, aq) = 0 \quad \Delta_f G^\ominus (H^+, aq) = 0$$



$$\Delta_r H^\ominus = -167.2 \text{ kJ mol}^{-1}$$

$$\Delta_r G^\ominus = -131.2 \text{ kJ mol}^{-1}$$

$$\Delta_r H^\ominus = \Delta_f H^\ominus (H^+, aq) + \Delta_f H^\ominus (Cl^-, aq)$$

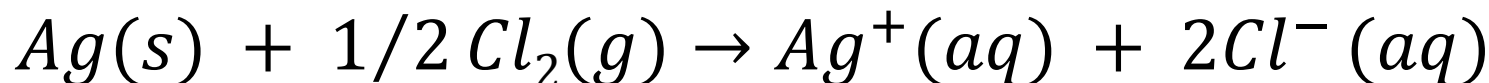
$$\rightarrow \Delta_f H^\ominus (Cl^-, aq) = -167.2 \text{ kJ mol}^{-1}$$

$$\Delta_r G^\ominus = \Delta_f G^\ominus (H^+, aq) + \Delta_f G^\ominus (Cl^-, aq)$$

$$\rightarrow \Delta_f G^\ominus (Cl^-, aq) = -131.2 \text{ kJ mol}^{-1}$$



With the value of  $\Delta_f G^\ominus (Cl^-, aq)$  established, we can find the value of  $\Delta_f G^\ominus (Ag^+, aq)$



$$\Delta_r H^\ominus = -61.62 \text{ kJ mol}^{-1}$$

$$\Delta_r G^\ominus = -54.12 \text{ kJ mol}^{-1}$$

$$\Delta_r H^\ominus = \Delta_f H^\ominus (Ag^+, aq) + \Delta_f H^\ominus (Cl^-, aq)$$

$$\rightarrow \Delta_f H^\ominus (Ag^+, aq) = +105.58 \text{ kJ mol}^{-1}$$

$$\Delta_r G^\ominus = \Delta_f G^\ominus (Ag^+, aq) + \Delta_f G^\ominus (Cl^-, aq)$$

$$\rightarrow \Delta_f G^\ominus (Ag^+, aq) = +77.11 \text{ kJ mol}^{-1}$$



## (b) Standard entropies of ions in solution

- Like  $G$  and  $H$  can't measure entropies of isolated ions in solution

Assume  $S^\ominus$  for  $H^+$  in solution at all temperatures is 0;

$$S^\ominus(H^+, aq) = 0$$

Derive values for other ions relative to it.

- Estimate of  $S^\ominus(H^+, aq)$  on absolute (3rd Law scale) is  $-21 J K^{-1} mol^{-1}$   
 $H^+$  induces order



- Some ions have positive  $S^\ominus$  and some negative relative to  $H^+$ :

$$S^\ominus(Cl^-, aq) = 57 J K^{-1} mol^{-1}$$

$$S^\ominus(Mg^{2+}, aq) = -128 J K^{-1} mol^{-1}$$

Means that relative to  $H^+$  in water  $Mg^{2+}$  induces more order and  $Cl^-$  less.

- They are related to the degree to which the ions order the water molecules around them in the solution:

Small highly charged ions ( $Mg^{2+}$ ) induce more order than bigger less charged ions





## 2. Ion activities

### (a) The definition of activity

Interactions between ions are so strong that the approximation of replacing activities by concentrations (molality) is valid only in very dilute solutions ( $< 10^{-3} \text{ mol kg}^{-1}$  in total ion concentration) and in precise work activities must be used.

$$\mu_B = \mu_B^\ominus(T) + RT \ln a$$

$$a = \gamma \frac{b}{b^\ominus} \quad \lim_{b \rightarrow 0} \gamma_B = 1 \quad \longrightarrow \quad a \rightarrow \frac{b}{b^\ominus}$$

ideal-dilute solution



$$\mu = \mu^{\ominus}(T) + RT \ln \gamma \frac{b}{b^{\ominus}}$$

$$\mu = \mu^{\ominus}(T) + RT \ln \frac{b}{b^{\ominus}} + RT \ln \gamma = \mu^{ideal} + RT \ln \gamma$$

$\mu^{ideal}$ : the chemical potential of the ideal-dilute solution of the same molality.



## (b) Mean activity coefficients

Consider a solution of two monovalent, cation ( $M^+$ ) and anion ( $X^-$ ); the total chemical potential of the ions in the electrically neutral solution is the sum of these partial molar quantities.

$$G = \mu_+ + \mu_- = \mu_+^{ideal} + \mu_-^{ideal} + RT \ln \gamma_+ + RT \ln \gamma_-$$

$$G^{ideal} = \mu_+^{ideal} + \mu_-^{ideal}$$

$$G = G^{ideal} + RT \ln \gamma_+ \gamma_-$$

All the deviations from ideality are in the last term.



There is no experimental way of separating the product  $\gamma_+\gamma_-$  into contributions from the cations and the anions. we introduce the mean activity coefficient  $\gamma_{\pm}$

For a 1,1-electrolyte:

$$\gamma_{\pm} = (\gamma_+\gamma_-)^{1/2}$$

Now the individual chemical potentials of the ions can be express as:

$$\begin{aligned}\mu_+ &= \mu_+^{ideal} + RT \ln \gamma_{\pm} \\ \mu_- &= \mu_-^{ideal} + RT \ln \gamma_{\pm}\end{aligned}$$



It can be generalized to the case of a compound  $M_pX_q$  that dissolves to give a solution of  $p$  cations and  $q$  anions, the mean activity coefficient

$$G = p\mu_+ + q\mu_- = G^{ideal} + pRT\ln\gamma_+ + qRT\ln\gamma_-$$

$$\gamma_{\pm} = (\gamma_+^p \gamma_-^q)^{1/s} \quad s = p + q$$

The chemical potential of each ion:

$$\mu_+ = \mu_+^{ideal} + RT\ln\gamma_{\pm}$$

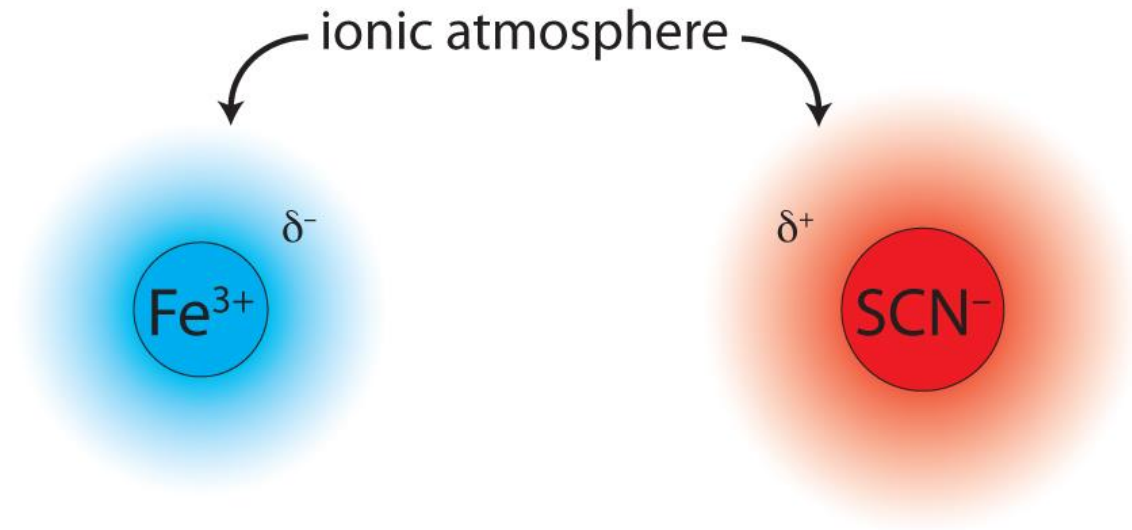
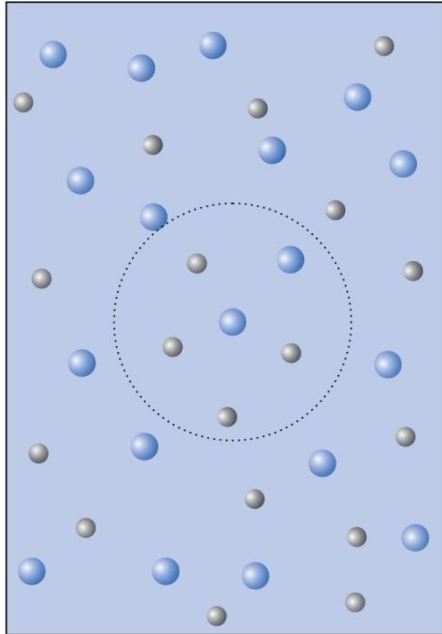
$$\mu_- = \mu_-^{ideal} + RT\ln\gamma_{\pm}$$

The total chemical potential of the ions in the electrically neutral solution:

$$G = p\mu_+ + q\mu_- = G^{ideal} + sRT\ln\gamma_{\pm}$$



## (c) The Debye-Hückel limiting law



**Ionic atmosphere:** The time-averaged, spherical haze around the central ion, in which counter ions outnumber ions of the same charge as the central ion, has a net charge equal in magnitude but opposite in sign to that on the central ion.



## The results of ionic atmosphere:

At very low concentrations, the  $\gamma_{\pm}$  can be calculated from the Debye-Hückel limiting law:

$$\log \gamma_{\pm} = -|z_+ z_-| A I^{1/2}$$

- $z_{\pm}$  is charged number on ions;
- Sign of charge included, e.g,  $z_{Na^+} = +1$ ;  $z_{SO_4^{2-}} = -2$ ;
- Works well at dilute solutions ( $b < 1$  mmol/kg).



$$\log \gamma_{\pm} = -|z_+ z_-| A I^{1/2}$$

$$A = \frac{F^3}{4\pi N_A \ln 10} \left( \frac{\rho b^{\ominus}}{2 \varepsilon^3 R^3 T^3} \right)^{1/2} \quad A = f(\text{solvent}, T)$$

$A = 0.509$  for an aqueous solution at 25°C

$$I = \frac{1}{2} \sum_i z_i^2 (b_i / b^{\ominus})$$

$I$ : ionic strength of the solution (dimensionless)

$z_i$ : the charge number of an ion  $i$  (+ for cations and – for anions)

$b_i$ : the ion's molality

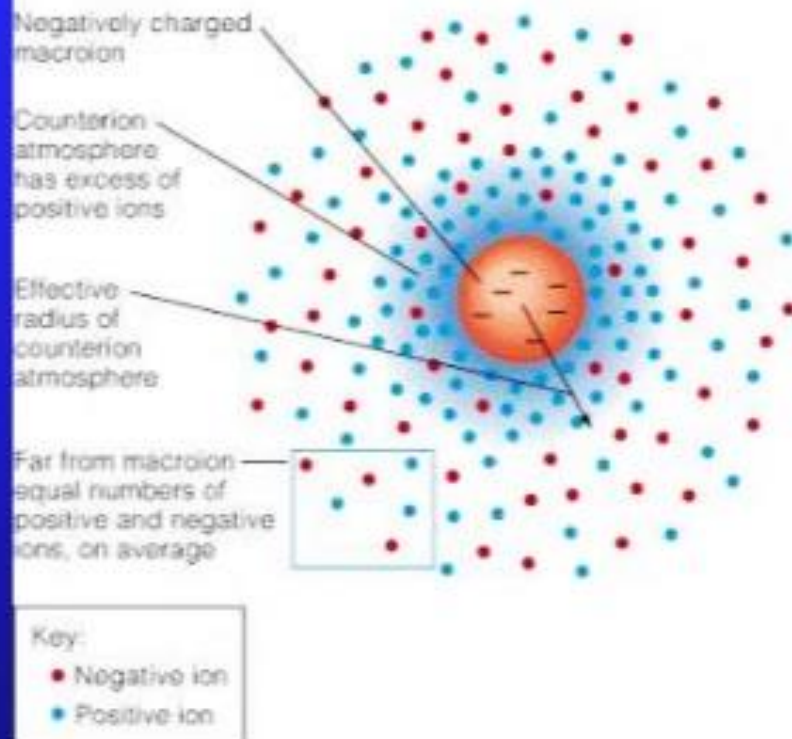
\* The sum extends over all the ions present in the solution.

For solutions consisting of two types of ion at molalities  $b_+$  and  $b_-$

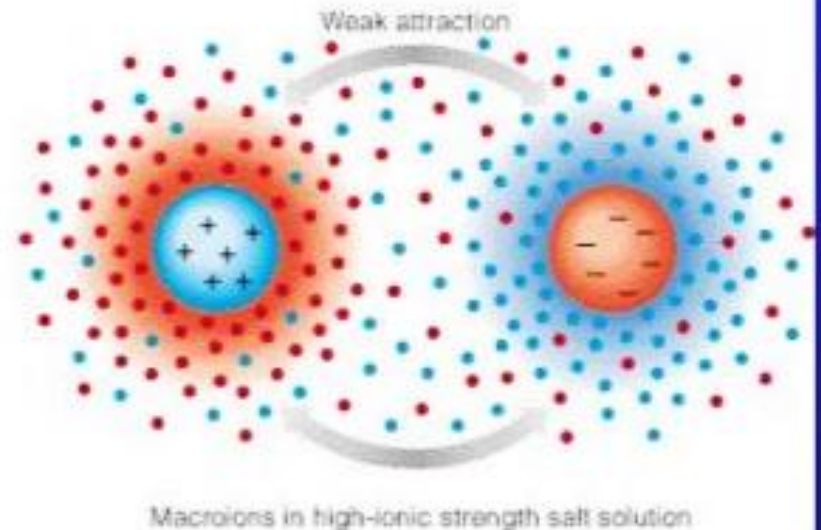
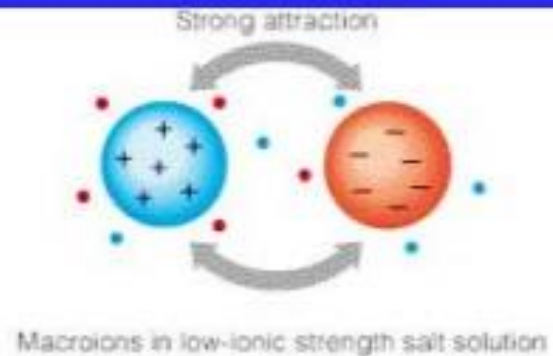
$$I = \frac{1}{2} (b_+ z_+^2 + b_- z_-^2) / b^{\ominus}$$



# The Effect of Ionic Strength



(a) The counterion atmosphere

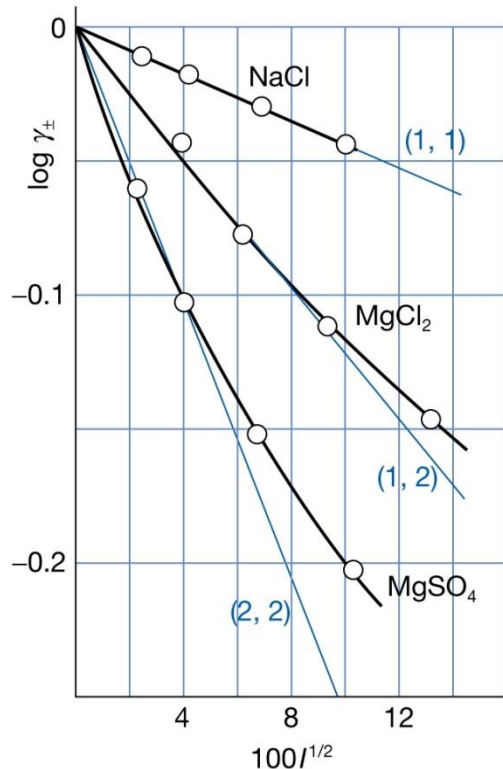


(b) The influence of ionic strength



# Debye-Hückel is a Limiting Law

$$\log \gamma_{\pm} = -|z_+ z_-| A I^{1/2}$$



*Molality* ↓

*good agreement*

$|z_+ z_-|$  ↓

$\gamma_{\pm}$  ↑

The departure from the ideal-dilute value is caused by the interaction of each ion with the ion atmosphere resulting from all other ions in the solution.

只有非常稀薄 (< 1 mmol/kg) 和电荷数比较少的情况下比较适用



## (d) The extended Debye-Hückel law

(1 mmol/kg < b < 0.1 mol/kg)

When the ionic strength of the solution is too high for limiting law to be valid, it is found that the activity coefficient may be estimated from the extended Debye-Huckel Law:

$$\log \gamma_{\pm} = - \frac{A |z_{+} z_{-}| I^{1/2}}{1 + B I^{1/2}}$$

B: dimensionless const., adjustable empirical parameter.

When b > 0.1 mol/kg

Model dependence of  $\gamma$  of solvent on solute and use Gibbs-Duhem equation ( $\sum n_i d\mu_i = 0$ ) to estimate  $\gamma$  of solute.



The mean activity coefficient of  $5.0 \times 10^{-3} \text{ mol kg}^{-1}$  KCl (aq) at 25°C is calculated by writing:

$$I = \frac{\frac{1}{2} (b_+ z_+^2 + b_- z_-^2)}{b^\ominus} = \frac{1}{2} (b_+ + b_-) / b^\ominus = b / b^\ominus$$

$$\begin{aligned} \log \gamma_{\pm} &= -|z_+ z_-| A I^{1/2} = -0.509 \times (5.0 \times 10^{-3})^{1/2} \\ &= -0.036 \end{aligned}$$

$$\gamma_{\pm} = 0.92$$



## 8.2 Electrochemical cells

### 1. Introduction

#### Electrochemical cell

Two electrodes in contact with an electrolyte

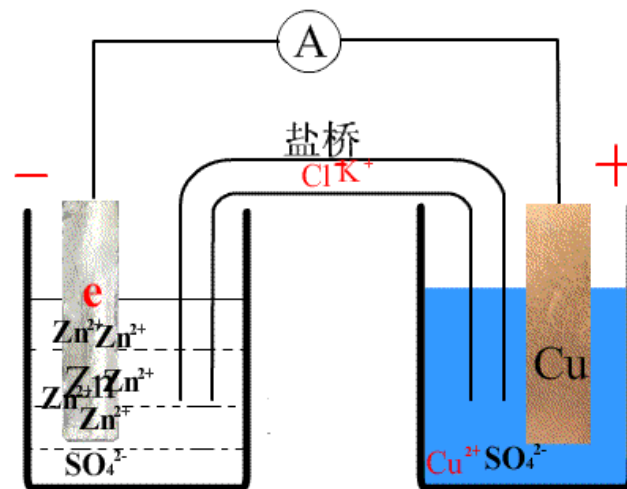
#### Electrolyte

An ionic conductor (solution, liquid, or solid)

#### Electrode compartment

electrode + electrolyte

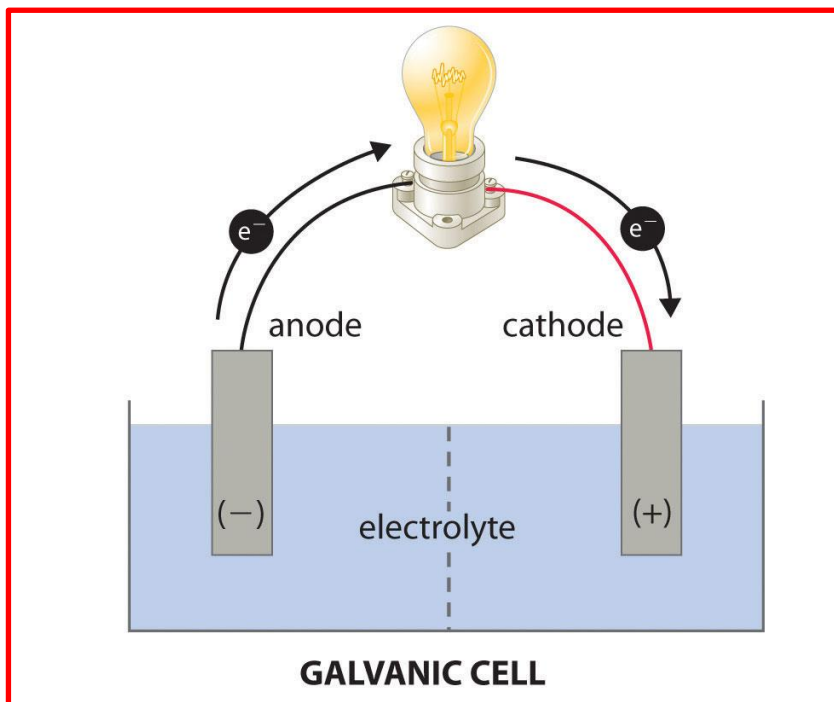
If electrolytes are different compartments may be connect with salt bridge (Electrolyte solution in agar jelly).



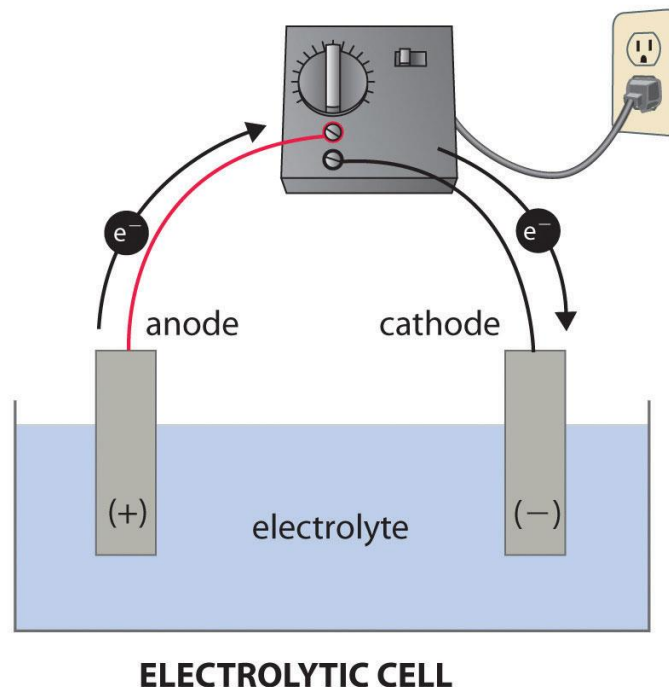


**Galvanic cell** - an electrochemical cell that produces electricity as result of the spontaneous reaction.

**Electrolytic cell** - an electrochemical cell in which a non-spontaneous reaction is driven by an external source of current.



Energy released by spontaneous redox reaction is converted to electrical energy.



Electrical energy is used to drive nonspontaneous redox reaction.



## 2. Half-reactions and electrodes

### a. Half-reactions

Redox reaction is one involving transfer of electrons

**OILRIG** (Oxidation Is Loss, Reduction Is Gain)

Oxidation is loss of electrons

Reduction is gain of electrons

Reducing agent (reductant) is electron donor

Oxidizing agent (oxidant) is electron acceptor



A **half reaction** is either the oxidation or reduction reaction component of a redox reaction. A half reaction is obtained by considering the change in oxidation states of individual substances involved in the redox reaction.

Any redox reaction can be expressed as the difference of two **reduction half reactions** (sum of oxidation and reduction half reaction)



The difference of the two (copper-zinc) is:







The oxidized and reduced substances in a half-reaction form a redox couple, denoted Ox/Red. The corresponding reduction half-reaction as:



The reaction quotient,  $Q$ , for the half-reaction, is defined like the reaction quotient for the overall reaction, but the electrons are ignored.

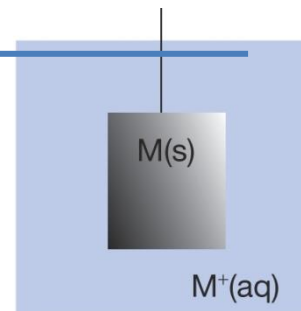


$$Q = \frac{1}{a_{\text{Cu}^{2+}}}$$



### Metal/metal ion (a)

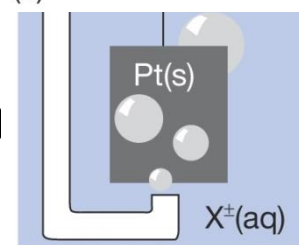
Designation:  $M(s)|M^+(aq)$   
 Redox couple:  $M^+/M$   
 Half reaction:  $M^+(aq) + 1e^- \rightarrow M(s)$



(a)

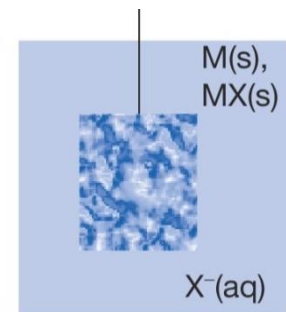
### Gas (b): \*Inert metal (Pt) source or sink of $e^-$

Designation\*:  $Pt(s)|X_2(g)|X^+(aq)$  or  $Pt(s)|X_2(g)|X^-(aq)$   
 Redox couple:  $X^+/X_2$  or  $X_2/X^-$   
 Half reaction:  
 $X^+(aq) + 1e^- \rightarrow 1/2X_2(g)$   
 or  $1/2X_2(g) + 1e^- \rightarrow X^-(aq)$



### Metal/insoluble salt (c)

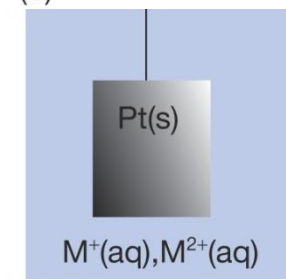
Designation:  $M(s)|MX(s)|X^-(aq)$   
 Redox couple:  $MX/M, X^-$   
 Half reaction:  $MX(s) + 1e^- \rightarrow M(s) + X^-(aq)$



(b)

### Redox (d):\*Inert metal (Pt) source or sink of $e^-$

Designation\*:  $Pt(s)|M^+(aq), M^{2+}(aq)$   
 Redox couple:  $M^{2+}/M^+$   
 Half reaction:  $M^{2+}(aq) + 1e^- \rightarrow M^+(aq)$



(d)



## b. Reactions at electrodes

In an electrochemical cell, the reduction and oxidation processes responsible for the overall reaction are separated in space: they take place in their own electrode compartment as the reaction proceeds, the electrons released in the oxidation at one electrode:



Traveling through the external circuit, the electrons re-enter the cell through the other electrode and bring about reduction:





## Anode (Anion)

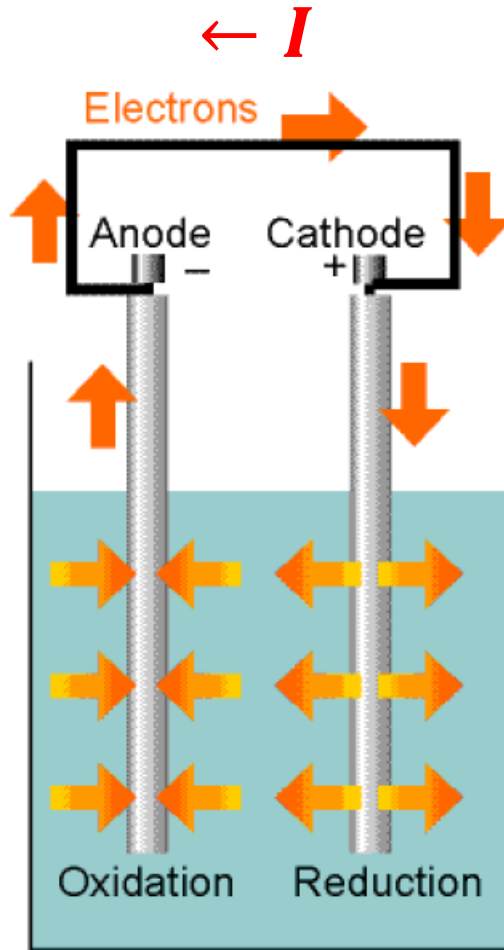
The electrode at which oxidation occurs

## Cathode (Cation)

The electrode at which reduction occurs.



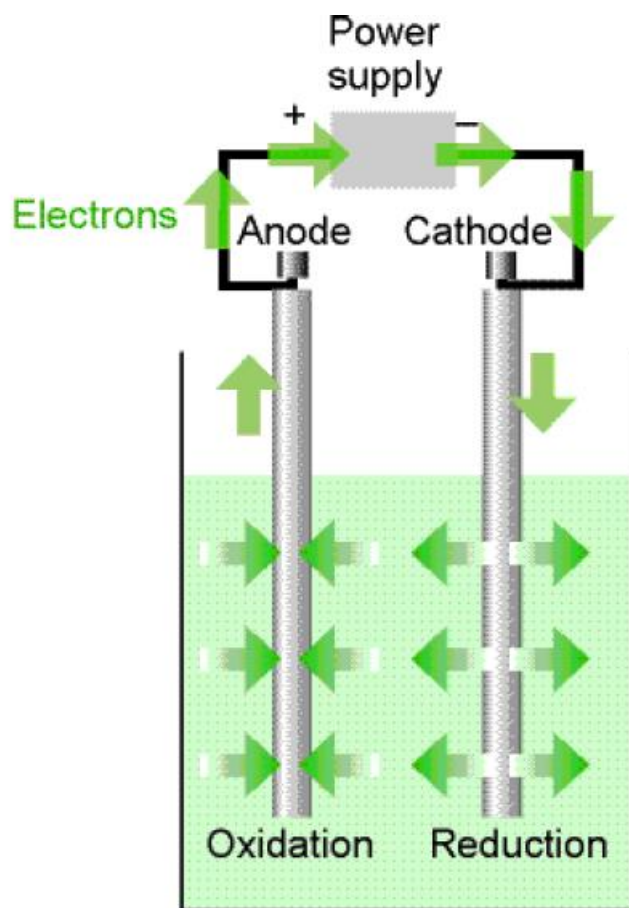
## Galvanic cell (produces electricity)



- ✓ Cathode at higher potential than anode.
- ✓ Species being reduced withdraws electrons from cathode giving it a relative positive charge.
- ✓ Species being oxidized deposits electrons in anode giving it a relative negative charge.



## Electrolytic cell (electricity supplied)



- ✓ Oxidation still occurs at anode
  - Oxidation doesn't occur spontaneously
  - Electrons come from the species in that compartment
  - Anode relatively positive to cathode
- ✓ Cathode
  - Supply of electrons drives reduction



### 3. Varieties of cells

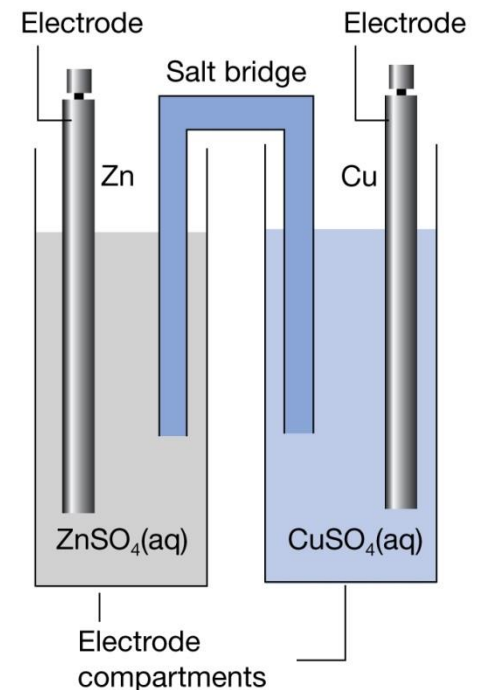
- Commonest cell has single electrolyte in contact with both electrodes.
- Daniel cell - electrode compartments separated.
- Concentration cell
  - **Electrolyte concentration cell** - same electrolyte, different concentration
  - **Electrode concentration cell** - electrodes have different concentration
    - e.g. Gas cells at different pressures
    - Amalgams at different concentrations



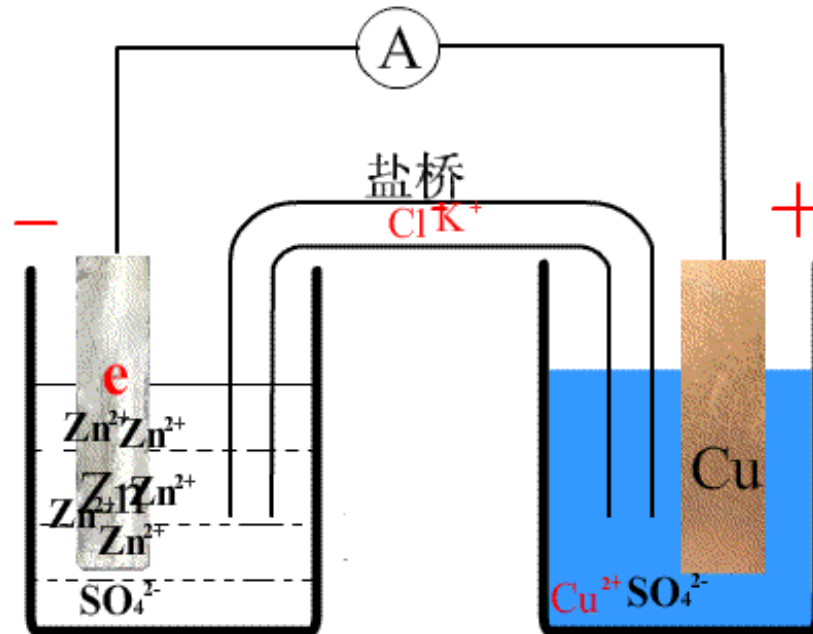
## a. Liquid junction potentials

In a cell with two different electrolyte solutions in contact, there is an additional source of potential difference, the liquid junction potential,  $E_{lj}$ , across the interface of the two electrolytes. Another liquid junction potential exist in the electrolyte concentrations cell.

- Due to differing mobility of ions of different sizes across interface.
- Can be reduced with salt bridge.  
Potential is then independent of concentration of electrolyte solution.







## Cell Notation Rules

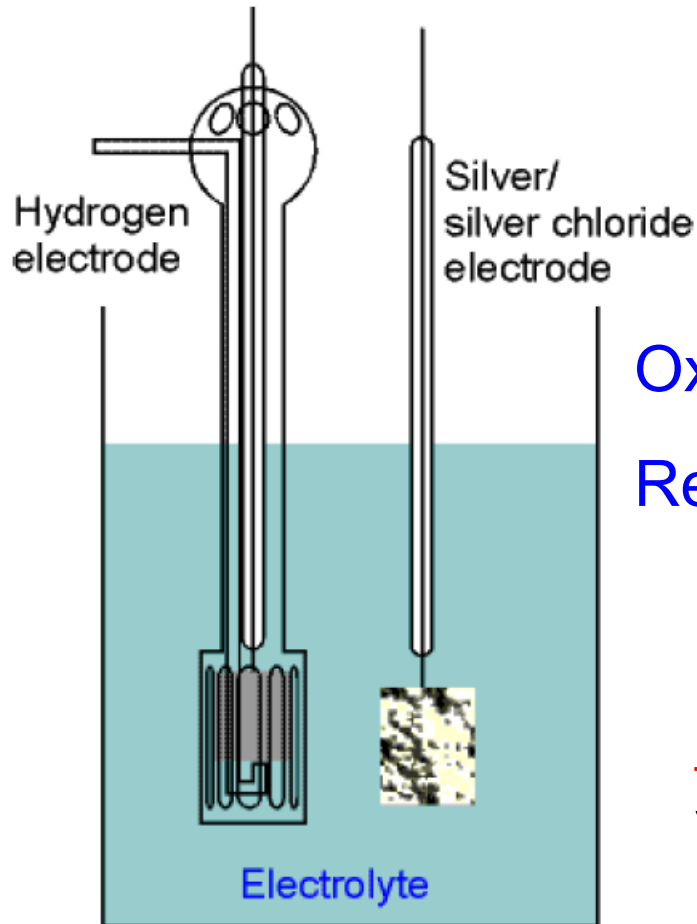
1. The anode half-cell is described first; the cathode half-cell follows. Within a given half-cell, the reactants are specified first and the products last. The description of the oxidation reaction is first, and the reduction reaction is last; when you read it, your eyes move in the direction of electron flow. Spectator ions are not included.
2. A single vertical line ( | ) is drawn between two chemical species that are in different phases but in physical contact with each other (e.g., solid electrode | liquid with electrolyte). A double vertical line ( || ) represents a salt bridge or porous membrane separating the individual half-cells.
3. The phase of each chemical (s, l, g, aq) is shown in parentheses. If the electrolytes in the cells are not at standard conditions, concentrations and/or pressure, they are included in parentheses with the phase notation. If no concentration or pressure is noted, the electrolytes in the cells are assumed to be at standard conditions (1.00 M or 1.00 atm and 298 K).

Using these rules, the notation for the cell we put together is:





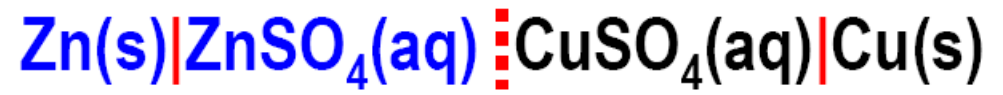
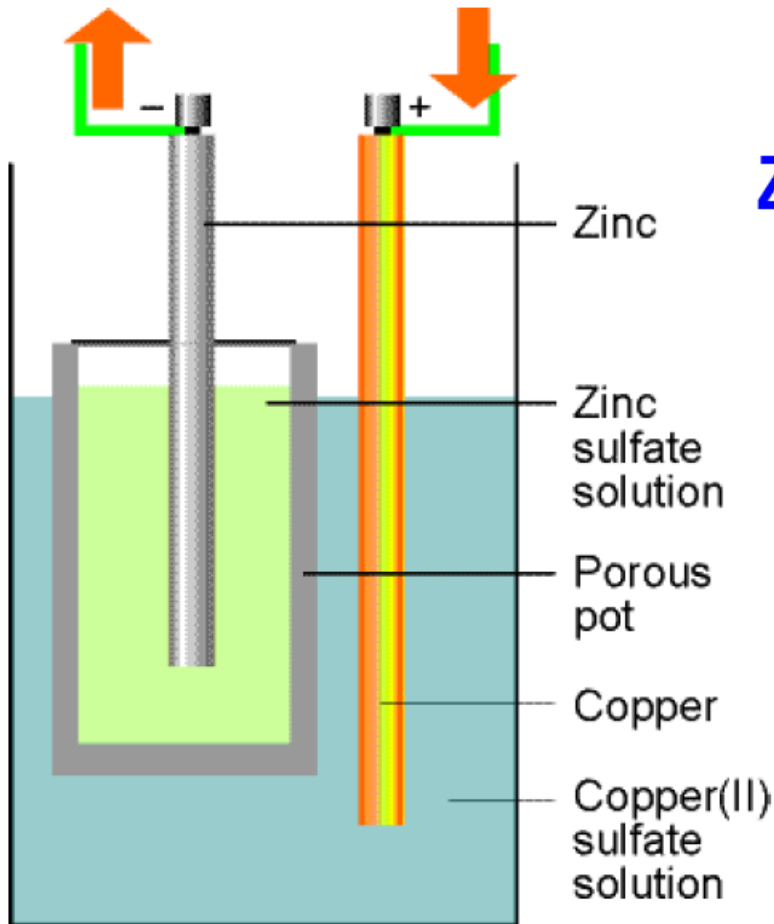
## b. Notation



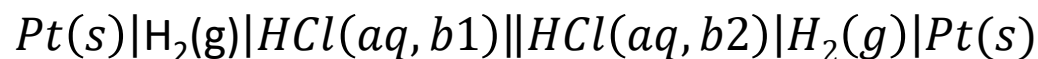
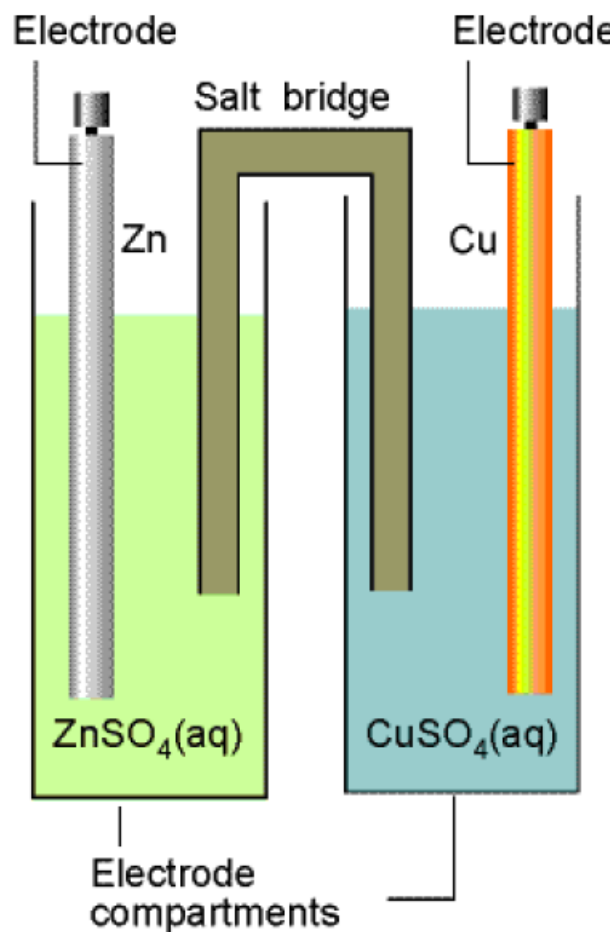
Oxidation reaction is at the left (Anode)

Reduction reaction is at the right (Cathode)

In the notation for cells, phase boundaries are denoted by a vertical bar (|)



A liquid junction is denoted by :



A double vertical line, **||**, denotes an interface for which it is assumed that the junction potential has been eliminated.

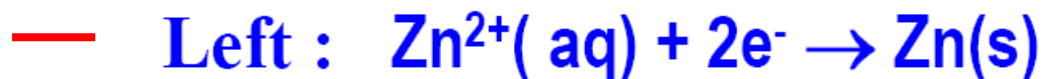
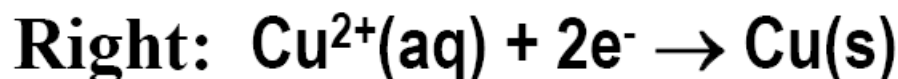


### c. The cell reaction

The current produced by a galvanic cell arises from the spontaneous chemical reaction taking place inside it. The cell reaction is written on the assumption that the right-hand electrode is the cathode, and hence that the spontaneous reaction is one in which reduction is taking place in the right-hand compartment; if it is, then the cell reaction is spontaneous as written. If the left-hand electrode turns out to be the cathode, then the reverse of the cell reaction is spontaneous.



the two electrodes' **reduction half-reactions** are



Subtract left from right



## d. The cell potential

A cell in which the overall cell reaction has not reached chemical equilibrium can do electrical work as the reaction drives electrons through an external circuit.

A cell in which the overall reaction is at equilibrium can do no work, and then the cell potential is zero.

The work that a given transfer of electrons can accomplish depends on the potential difference between the two electrodes. This potential difference is called the cell potential and is measured in volts, V.





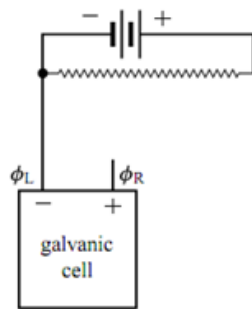
The maximum electrical work that a system can do is given by the value of  $\Delta G$ , for a spontaneous process at constant temperature and pressure:

$$w_{e,max} = \Delta G$$

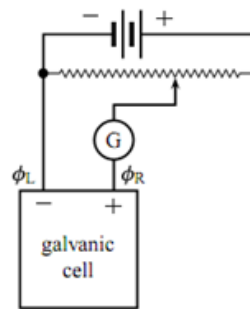
To make thermodynamic measurements on the cell by measuring the work it can do, we must ensure that it is operating reversibly. Only then it is producing maximum work and can be relate that work to  $\Delta G$ ;



Moreover, to measure  $\Delta_r G$  we must ensure that the cell is operating reversibly at a specific, constant composition. Both these conditions are achieved by measuring the cell potential when it is balanced by an exactly opposing source of potential so that the cell reaction occurs reversibly and the composition is constant. The resulting potential difference is called the zero-current cell potential,  $E$  (formerly, and still commonly, the 'electromotive force', or emf, of the cell).



(a)



(b)



## e. The relation between $E$ and $\Delta_r G$

The relation between the reaction Gibbs energy and the zero-current cell potential is:

$$- \nu F E = \Delta_r G$$

$F$ : Faraday constant,

$F = eN_A$ , the magnitude of the charge per mole of electrons.

$\nu$ : The number of electrons transferred

The connection between electrical measurements and thermodynamic properties



$$\Delta_r G = \left( \frac{\partial G}{\partial \xi} \right)_{T,p}$$

$$dG = \Delta_r G \times d\xi = dw_e$$

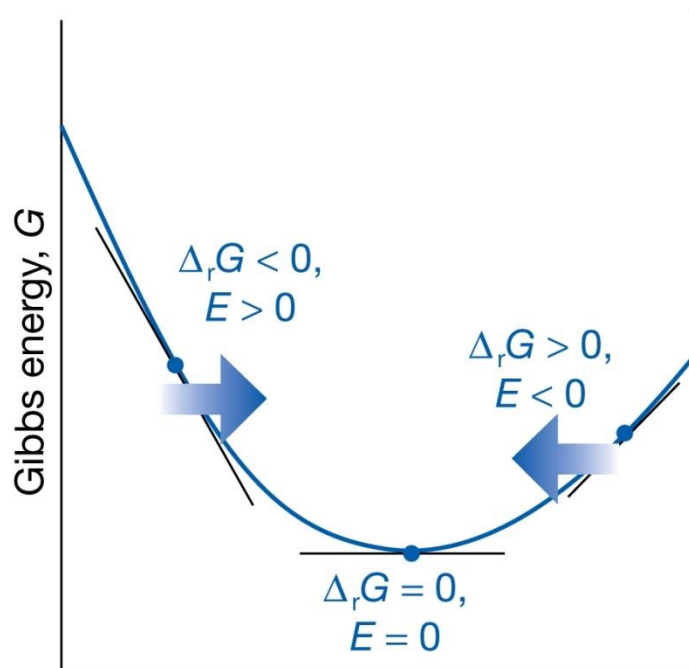
$$dw_e = -eN_A \times v d\xi \times E = -F \times v d\xi \times E$$

$-eN_A = F$ : the charge per mole of electrons

$v d\xi$ : the amount of electrons

$E$ : emf

$$\Delta_r G = -vFE$$



$$- \nu F E = \Delta_r G$$

$E > 0$ , the slope is steep, far from equilibrium,  
spontaneous reaction;

$E \rightarrow 0$ , the slope is close to zero;

$E = 0$ , the slope is zero,  $\Delta_r G = 0$ ;

$E < 0$ , the reverse reaction is spontaneous



## f. The Nernst equation

$$\Delta_r G = \Delta_r G^\ominus + RT \ln Q$$

$$-vFE = \Delta_r G$$



$$E = -\frac{\Delta_r G^\ominus}{vF} - \frac{RT}{vF} \ln Q$$

Because  $RT/F = 25.7 \text{ mV}$  at  $25^\circ\text{C}$

$$E = E^\ominus - \frac{25.7 \text{ mV}}{v} \ln Q$$

$$E^\ominus = -\frac{\Delta_r G^\ominus}{vF}$$

Standard emf of the cell



$$E = E^\ominus - \frac{RT}{vF} \ln Q$$

Nernst equation



$E^\ominus$  is the emf when all reactants and products are in their standard states.

$$a_{\text{products}} = 1$$

$$a_{\text{reactants}} = 1,$$

$$\text{so } Q = 1 \text{ and } \ln(Q) = 0$$

The standard emf  $E^\ominus$  is a disguised form of the standard reaction Gibbs energy

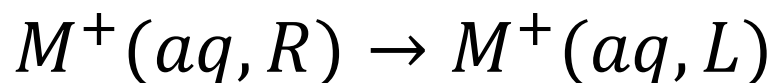


## g. Concentration cells

For the potential of an electrolyte concentration cell



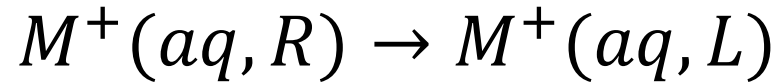
where the solutions L and R have different molalities. The cell reaction is:



$$Q = \frac{a_L}{a_R} \quad v = 1$$

$E^\ominus$  is 0 because when  $R=L$  the two compartments are identical and no driving force.





$$E = -\frac{RT}{F} \ln \frac{a_L}{a_R} \approx -\frac{RT}{F} \ln \frac{b_L}{b_R}$$

- If  $R > L$ ,  $\ln(b_L/b_R) < 0$ ,  $E > 0$ ,  $\Delta_r G < 0$  concentration will be lowered by reduction in right compartment.
- If  $L > R$ ,  $\ln(b_L/b_R) > 0$ ,  $E < 0$ ,  $\Delta_r G > 0$  concentration will be lowered by reduction in left compartment.
- This has biological application - nerve firing involves change in permeability of cell membrane to  $\text{Na}^+$   
This changes nerve cell potential.





## h. Cells at equilibrium

A special case of Nernst equation has great importance in electrochemistry. Suppose the reaction has reached equilibrium, then  $Q=K$ . A chemical reaction at equilibrium cannot do work, and hence it generates zero potential difference between the electrodes of a galvanic cell.

If set  $E=0$  and  $Q=K$  in the Nernst equation it gives

$$E = E^{\ominus} - \frac{RT}{vF} \ln Q \quad \longrightarrow \quad \ln K = \frac{vF E^{\ominus}}{RT}$$



## 4. Standard potentials

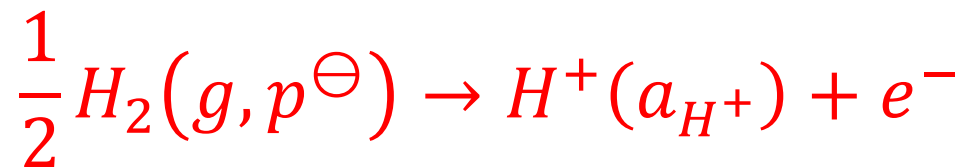
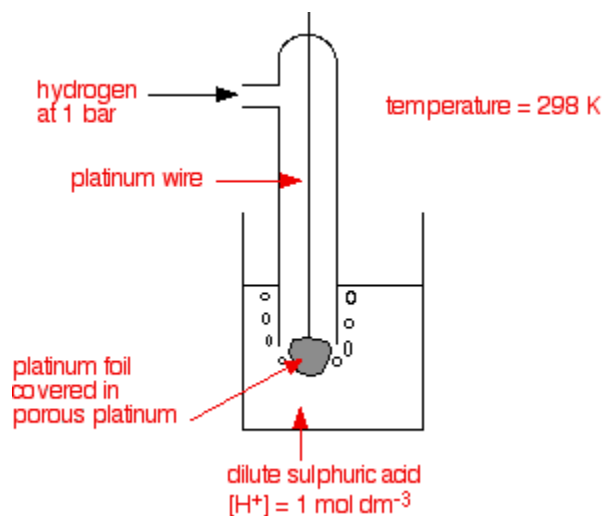
A galvanic cell is a combination of two electrodes, and each one can be considered as making a characteristic contribution to the overall cell potential. Although it is not possible to measure the contribution of a single electrode, we can define the potential of one of the electrodes as having a zero potential and then assign values to others on that basis. The specially selected electrode is the standard hydrogen electrode (SHE):



at all temperatures:  $E^\ominus = 0$



## Standard hydrogen electrode(SHE)



$$b_{H^+} = 1 \text{ mol kg}^{-1}; \quad \gamma_{H^+} = 1; \quad a_{H^+} = 1$$

$$E^\ominus\{H^+|H_2(g)\} = 0$$



The standard potential,  $E^\ominus$ , of another couple is then assigned by constructing a cell in which it is the right-hand electrode and the standard hydrogen electrode is the left-hand electrode.

$$E^\ominus = E^\ominus(\text{right}) - E^\ominus(\text{left})$$

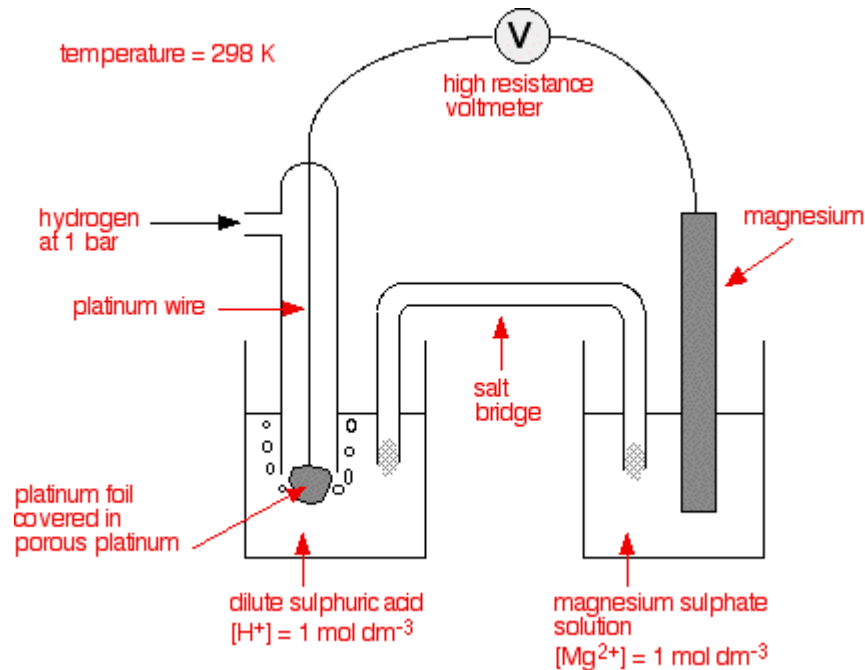
**The standard emf of cells**

**The standard potential of electrode**



# Using the SHE to measure electrode potential

In the actual experimental setup the two half cells are connected together via an external circuit wire and a salt bridge to make the whole cell.

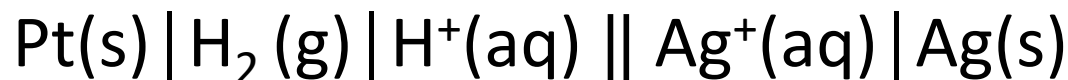


*SHE || a given electrode*

$$E^{\ominus} = E^{\ominus}(\text{right}) - E^{\ominus}(\text{left})$$



$E^\ominus$  attentions



$$E^\ominus (\text{Ag}^+ / \text{Ag}) = +0.80 \text{ V}$$



$$E^\ominus (\text{AgCl} / \text{Ag}, \text{Cl}^-) = +0.22 \text{ V}$$

### The standard potential of electrode

The standard potential is determined by properties of the hydrogen electrode as well as the couple to which the potential refers.





## $E^\ominus$ attentions

### Numerical factors

If standard emf reaction is multiplied by numerical factor,  $\Delta_r G$  increases by that factor

### Standard potential does not increase!

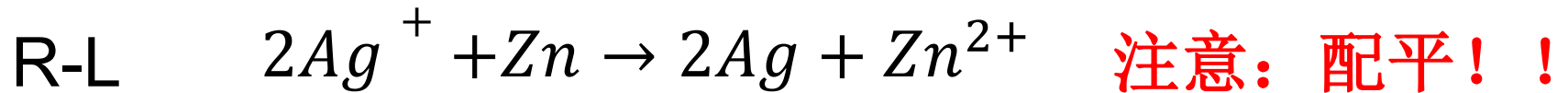
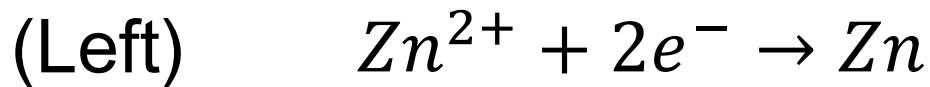
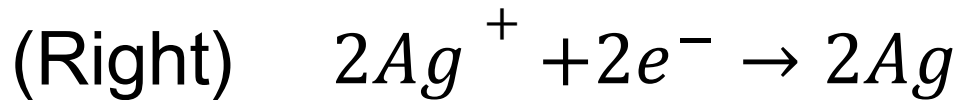
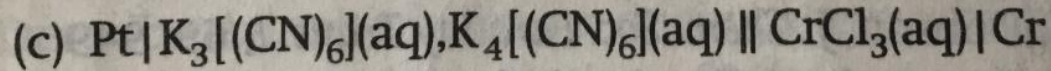
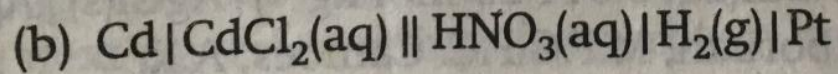
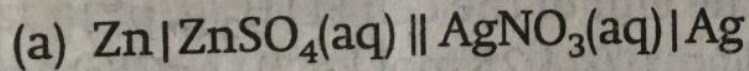
Recall  $E^\ominus = -\frac{\Delta_r G^\ominus}{vF}$

If  $\Delta_r G^\ominus(\text{new}) = n \times \Delta_r G^\ominus$ ,  $v(\text{new}) = n \times v$

$$E^\ominus(\text{new}) = -\left(\frac{\Delta_r G^\ominus(\text{new})}{v(\text{new})F}\right) = -\left(\frac{n \times \Delta_r G^\ominus}{n \times v F}\right) = E^\ominus$$



**(10.18 (a))** Write the cell reactions and electrode half-reactions for the following cells:



$$E = E^\ominus - \frac{RT}{vF} \ln Q \quad E^\ominus = E^\ominus(\text{right}) - E^\ominus(\text{left})$$

- $E^\ominus$  查表出来是多少就是多少，跟方程式写法无关。
- 求整体的 emf of cell，要写出整体的氧化还原反应式，这时要配平， $v$  and  $Q$  都和方程式的写法想关，要保持一致。

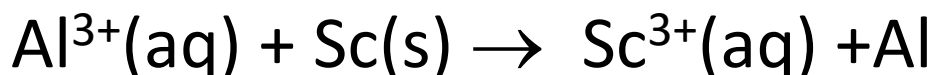
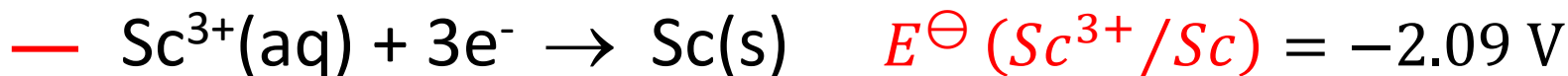


## a. The cell emf in terms of individual standard potentials

Cell emf of any cell can be calculated from table of standard potentials. The emf of cell is just difference in standard potential of electrodes.

1) Write new cell diagram

$$2) E^{\ominus}(\text{overall}) = E^{\ominus}(\text{right}) - E^{\ominus}(\text{left})$$



$$E^{\ominus} = E^{\ominus}(\text{Al}^{3+}/\text{Al}) - E^{\ominus}(\text{Sc}^{3+}/\text{Sc})$$

$$= 1.69 \text{ V} - (-2.09 \text{ V}) = 3.78 \text{ V}$$



Recall:  $E^\ominus = -\frac{\Delta_r G^\ominus}{vF}$        $\ln K = \frac{vFE^\ominus}{RT}$

If  $E^\ominus > 0$ ,  $\Delta_r G^\ominus < 0$  and  $K > 1$

Example above, at 25°C,

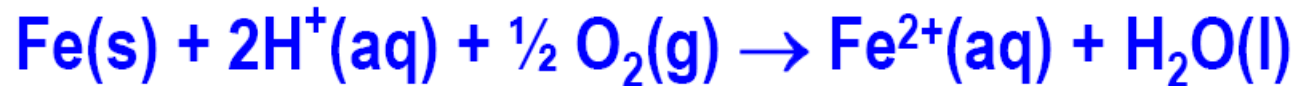
$$K = \exp(vFE^\ominus / RT) = 4.27 \times 10^{191}$$

$$K = [\text{Sc}^{3+}] / [\text{Al}^{3+}]$$



**Identify the spontaneous direction of a reaction**

**One of the reactions important in corrosion in an acidic environment is**



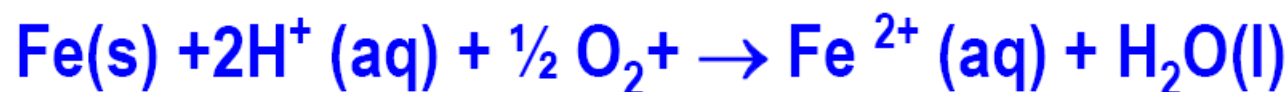
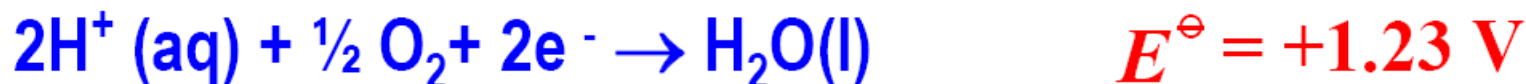
**Does the equilibrium constant favor the formation of  $\text{Fe}^{2+}(\text{aq})$ .**



**Method:** We need to decide whether the standard potential for the reaction as written is positive, for a positive value would imply that  $\Delta_r G^\ominus < 0$  and hence that  $K > 1$ . The sign of the cell potential is found by identifying the half-reactions that make up the overall reaction, and then taking their standard potentials from the table.



**Answer:** The two reduction half-reactions are



$$E^\ominus = +1.67 \text{ V}$$

Because  $E^\ominus > 0$  ,  $K > 1$ , the reaction favor products.



## Calculating an equilibrium constant

Calculate the equilibrium constant for the disproportionation  $2\text{Cu}^+(\text{aq}) \rightarrow \text{Cu}(\text{s}) + \text{Cu}^{2+}(\text{aq})$  at 298.15 K.

**Method:** The strategy is to calculate the standard potential for the cell in which the reaction of interest is the cell reaction. To proceed, express the overall reaction as the difference of two reduction half-reactions and then find the corresponding standard potentials by referring to the Table.

Use  $RT/F = 0.025693 \text{ V}$ .





**Answer:** The half-reactions and standard potentials we require are



The standard cell potential is therefore

$$E^\ominus = 0.52\text{V} - 0.16\text{V} = 0.36\text{V}$$

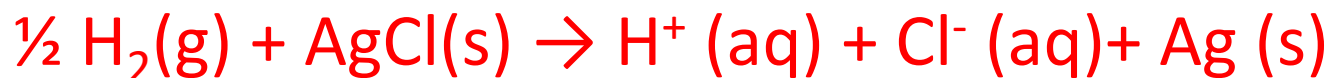
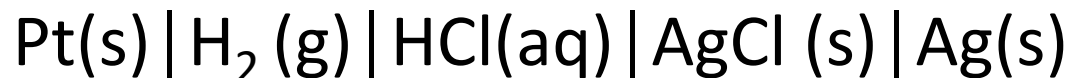
Then, because  $\nu = 1$ ,

$$\ln K = \frac{\nu F E^\ominus}{RT} = \frac{0.36 \text{ V}}{0.025693 \text{ V}} = 14 \quad K = 1.2 \times 10^6$$



## b. The measurement of standard potentials (Harned cell)

The procedure for measuring a standard potential is illustrated by considering the silver chloride electrode:



$$E = E^\ominus (\text{AgCl/Ag}, \text{Cl}^-) - \frac{RT}{F} \ln \frac{a_{\text{H}^+} \times a_{\text{Cl}^-}}{(f_{\text{H}_2}/p^\ominus)^{1/2}}$$

$$f_{\text{H}_2} = p^\ominus$$

$$E = E^\ominus (\text{AgCl/Ag}, \text{Cl}^-) - \frac{RT}{F} \ln(a_{\text{H}^+} \times a_{\text{Cl}^-})$$



$$E = E^{\ominus} - \frac{RT}{F} \ln(a_{H^+} \times a_{Cl^-})$$

$$a_{H^+} \times a_{Cl^-} = (\gamma_{\pm} \times b)^2$$

$\gamma_{\pm}$  is unknown

$$E = E^{\ominus} - \frac{RT}{F} \ln b^2 - \frac{RT}{F} \ln \gamma_{\pm}^2$$

$$E + \frac{2RT}{F} \ln b = E^{\ominus} - \frac{2RT}{F} \ln \gamma_{\pm}$$

$$\log \gamma_{\pm} = -|z_+ z_-| A I^{1/2}$$

$$I = \frac{1}{2} \sum_i z_i^2 (b_i / b^{\ominus}) = \frac{1}{2} (b_+ z_+^2 + b_- z_-^2) / b^{\ominus}$$



$$\ln \gamma_{\pm} \propto -b^{1/2}$$



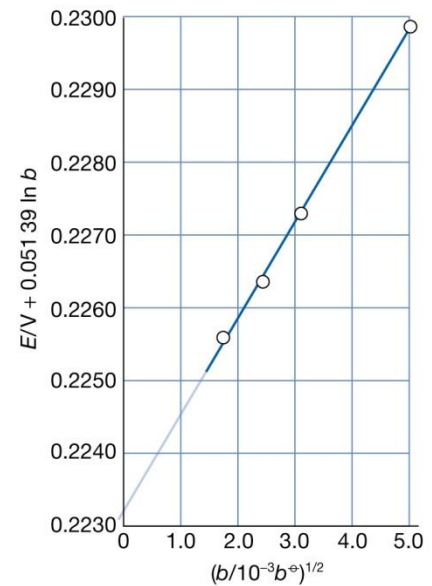
$$E + \frac{2RT}{F} \ln b = E^{\ominus} - \frac{2RT}{F} \ln \gamma_{\pm}$$

If the constant of proportionality in this relation written as:

$$\ln \gamma_{\pm} = \frac{F}{2RT} C (-b^{1/2})$$

$$E + \frac{2RT}{F} \ln b = E^{\ominus} + C b^{1/2}$$

This means a plot of  $E + \frac{2RT}{F} \ln b$  vs  $b^{1/2}$  has  $E^{\ominus}$  as intercept





### c. The measurement of activity coefficient

Once the standard potential of an electrode in a cell is known, the activities of the ions with respect to which it is reversible can be determined simply by measuring the cell potential with the ions at the concentration of interest.

$$E + \frac{2RT}{F} \ln b = E^{\ominus} - \frac{2RT}{F} \ln \gamma_{\pm}$$

$$\ln \gamma_{\pm} = \frac{E^{\ominus} - E}{2RT/F} - \ln b$$



## 8.3 Application of standard potential

### Calculate thermodynamic functions

(a) The reaction Gibbs energy

$$\Delta_r G = -vFE$$

$$\Delta_r G^\ominus = -vFE^\ominus$$

(b) The temperature coefficient of the cell potential

$$\left(\frac{\partial \Delta_r G}{\partial T}\right)_p = -\Delta_r S \qquad \Delta_r S = vF \left(\frac{\partial E}{\partial T}\right)_p$$

$$\frac{dE}{dT} = \frac{\Delta_r S}{vF}$$

$$\frac{dE^\ominus}{dT} = \frac{\Delta_r S^\ominus}{vF}$$

$\partial E / \partial T$ : The temperature coefficient of the cell potential



(c) The standard reaction enthalpy

$$\Delta_r G^\ominus = \Delta_r H^\ominus - T \Delta_r S^\ominus$$

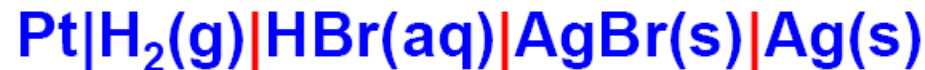
$$\Delta_r H^\ominus = -vFE^\ominus + vFT \frac{dE^\ominus}{dT}$$

This expression provides a non-calorimetric method for measuring  $\Delta_r H^\ominus$  and, through the convention  $\Delta_f H^\ominus (H^+ aq) = 0$  the standard enthalpies of formation of ions in solution. Thus, electrical measurements can be used to calculate all the thermodynamic properties of ions in solutions.



## Using the temperature coefficient of the cell potential

The standard cell potential of



was measured over a range of temperatures, and the data were fitted to the following polynomial:

$$E^\ominus/\text{V} = 0.07131 - 4.99 \times 10^{-4}(T/\text{K} - 298) - 3.45 \times 10^{-6}(T/\text{K} - 298)^2$$

Evaluate the standard reaction Gibbs energy, enthalpy, and entropy at 298 K.





**Method:** The **standard Gibbs energy** of reaction is obtained by using the following equation after evaluating  $E^\ominus$  at 298 K.

$$E^\ominus = -\frac{\Delta_r G^\ominus}{\nu F}$$

The **standard entropy** of reaction is obtained by using the following equation, which involves differentiating the polynomial with respect to  $T$  and then setting  $T = 298$  K.

$$\frac{dE^\ominus}{dT} = \frac{\Delta_r S^\ominus}{\nu F}$$

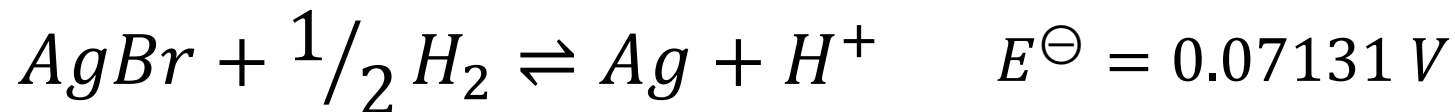
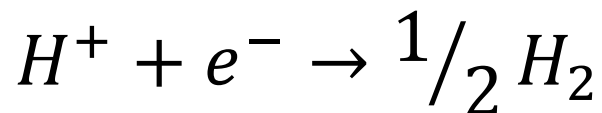
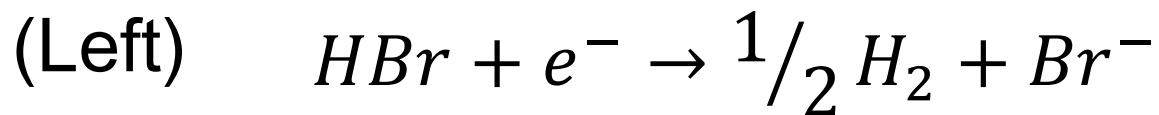
The **reaction enthalpy** is obtained by combining the values of the standard Gibbs energy and entropy.



$$E^{\ominus} = 0.07131 - 4.99 \times 10^{-4}(T - 298) - 3.45 \times 10^{-6}(T - 298)^2$$

$$T = 298 \text{ K}$$

$$E^{\ominus} = 0.07131 \text{ V}$$





$$\begin{aligned}\Delta_r G^\ominus &= -vFE^\ominus = -1 \times 9.65 \times 10^4 \text{ A} \cdot \text{s} \cdot \text{mol}^{-1} \times 0.07131 \\ &= -6.880 \times 10^3 \text{ Jmol}^{-1}\end{aligned}$$

$$E^\ominus = 0.07131 - 4.99 \times 10^{-4}(T - 298) - 3.45 \times 10^{-6}(T - 298)^2$$

$$\frac{dE^\ominus}{dT} = -4.99 \times 10^{-4} - 2 \times 3.45 \times 10^{-6}(T - 298) = -4.99 \times 10^{-4}$$

$$\frac{dE^\ominus}{dT} = \frac{\Delta_r S^\ominus}{vF}$$

$$\begin{aligned}\Delta_r S^\ominus &= vF \frac{dE^\ominus}{dT} = 1 \times 9.65 \times 10^4 \text{ A} \cdot \text{s} \cdot \text{mol}^{-1} \times (-4.99 \times 10^{-4}) \\ &= -48.2 \text{ J}K^{-1} \text{ mol}^{-1}\end{aligned}$$

$$\begin{aligned}\Delta_r H^\ominus &= \Delta_r G^\ominus + T\Delta_r S^\ominus \\ &= -2.12 \times 10^4 \text{ Jmol}^{-1}\end{aligned}$$



## Summary

### 1. Standard functions of formation of ions

$$\Delta_f H^\ominus(H^+, aq) = 0 \quad \Delta_f G^\ominus(H^+, aq) = 0$$

$$S^\ominus(H^+, aq) = 0$$

### 2. Mean activity coefficients (Debye-Hückel limiting law)

$$\log \gamma_{\pm} = -|z_+ z_-| A I^{1/2} \quad I = \frac{1}{2} \sum_i z_i^2 (b_i / b^\ominus)$$

$$A = 0.509 \text{ for water medium at } 298K$$



From  $\gamma_{\pm}$  to  $\mu$ :

$$\gamma_{\pm} = (\gamma_+^p \gamma_-^q)^{1/s} \quad s = p + q$$

The chemical potential of each ion:

$$\mu_+ = \mu_+^{ideal} + RT \ln \gamma_{\pm}$$

$$\mu_- = \mu_-^{ideal} + RT \ln \gamma_{\pm}$$

The total chemical potential of the ions in the electrically neutral solution:

$$\mu = p\mu_+ + q\mu_- = \mu^{ideal} + sRT \ln \gamma_{\pm}$$

Something you should know but not necessarily follow:

The real results show perfect agreement with the Debye-Hückel limiting law when the molality of the electrolytes less than about 1 mmol/Kg; There is an extended Debye-Hückel law which works for the solutions up to about 0.1 mol/kg.

However, only the Debye-Hückel limiting law will be tested.



### 3. The bridge between electrochemical reactions and thermodynamics.

Constant T and p

$$\Delta_r G = -vFE$$

$$\Delta_r G^\ominus = -vFE^\ominus \quad (\ominus: b_{ion} = 1 \text{ mol/kg})$$

$$E^\ominus = f(T)$$

E: electromotive force, emf (dimension is V).

The ability of galvanic cells to convert the chemical energy to the electrical energy.

We must ensure that the cell is operating reversibly at a specific, constant composition.



At constant temperature and pressure:

$$\Delta G \leq w_{add}$$

= *Reversible*

< *Spontaneous*



$$(dG)_{T,p} = \sum_{i=1}^k \mu_i dn_i = \sum_{i=1}^k \mu_i \nu_i d\xi$$

$$\left(\frac{\partial G}{\partial \xi}\right)_{T,p} = \sum_{i=1}^k \nu_i \mu_i = \Delta_r G$$

$\nu_i$ : Stoichiometry number

$\xi$ : Extent of reaction; The dimensions is amount of substance (moles).

$\Delta_r G$ : The reaction Gibbs energy, is defined as the slope of the graph of the Gibbs energy plotted against the extent of reaction,  $\xi$ . Dimension ( $\text{J mol}^{-1}$ ).





## 4. Nernst equation

$$E = E^{\ominus} - \frac{RT}{vF} \ln Q$$

$$emf = f(T, p, composition)$$

$emf = 0$  the cell reach its chemical equilibrium



## 5. Half reaction

The right-hand electrode: Cathode (Reduction reaction)

The left-hand electrode: Anode (Oxidation reaction)

How to design a galvanic cell?

Compare the reduction ability between two electrodes



$$E = E(\text{right}) - E(\text{left})$$

$$E^\ominus = E^\ominus(\text{right}) - E^\ominus(\text{left})$$

$$E^\ominus\{H^+(aq, b^\ominus) | H_2(g, p^\ominus)\} = 0$$



$$E^{\ominus}$$

- The reduction potential of an electrode (a redox couple)
- If a half reaction is multiplied by numerical factor,  $E^{\ominus}$  is not affected.
- The values of common electrodes can be obtained from information table 10.7.

$$\ln K = \frac{vFE^{\ominus}}{RT} \quad \text{Calculate equilibrium constant}$$



## 6. Calculate thermodynamic functions

$$\Delta_r G^\ominus = -vFE^\ominus$$

$$\frac{dE^\ominus}{dT} = \frac{\Delta_r S^\ominus}{vF}$$

$$\Delta_r G^\ominus = \Delta_r H^\ominus - T\Delta_r S^\ominus$$